1.3 Vector Equations

- A matrix with only one column is called a column vector, or simply a vector.

Ex:

$u = \begin{bmatrix} 3 \\ -1 \end{bmatrix}_{2\times 1}$ $v = \begin{bmatrix} 0.2 \\ 0.3 \end{bmatrix}$ $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$

where $w_1, w_2 \in \mathbb{R}$

where $\mathbb{R}$ is the set of real numbers, i.e $\mathbb{R} = (-\infty, \infty)$

- The set of all vectors with two entries is denoted by $\mathbb{R}^2$. (read r-two)
- The vectors in $\mathbb{R}^2$ are equal if and only if (iff) their corresponding values are equal.
- Given two vectors u and v in $\mathbb{R}^2$ their sum is the vector $u + v$ obtained by adding corresponding entries of u and v. For example:

$$\begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

- Given a vector u and a real number c, the scalar multiple of u by c is the vector cu obtained by multiplying each entry in u by c. For example:

for $u = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ and $c = 5$, then $cu = 5\begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 15 \\ -5 \end{bmatrix}$

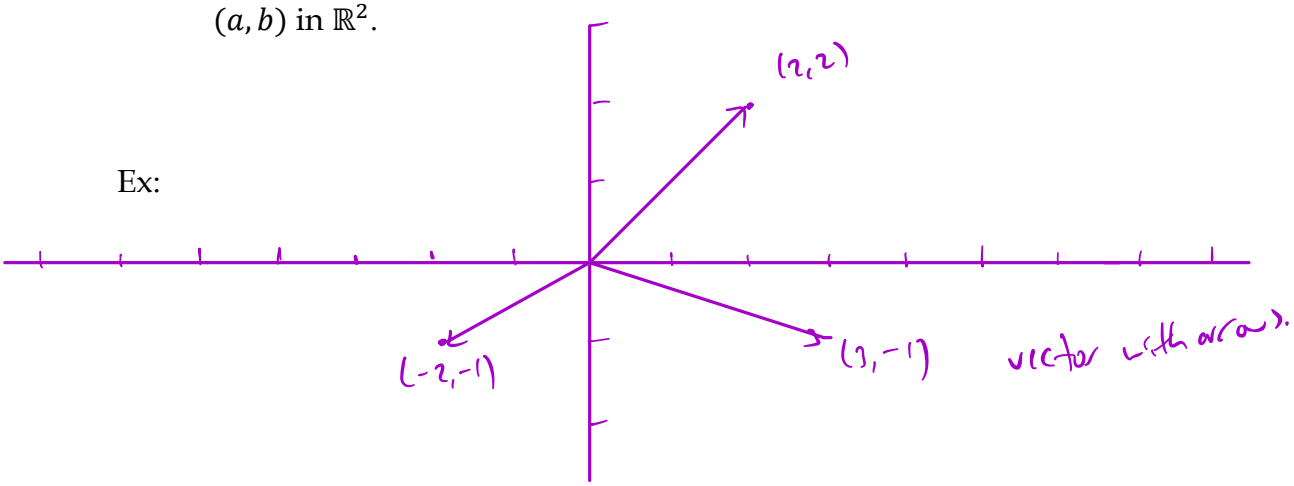
Ex: for $u = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$ find $4u$, $(-3)v$ and $4u + (-3)v$.

$4u = \begin{bmatrix} 4 \\ -8 \end{bmatrix}$, $(-3)v = \begin{bmatrix} -6 \\ 15 \end{bmatrix}$, $4u + (-3)v = \begin{bmatrix} 4 \\ -8 \end{bmatrix} + \begin{bmatrix} -6 \\ 15 \end{bmatrix} = \begin{bmatrix} -2 \\ 7 \end{bmatrix}$

– Sometimes for convenience, we can write a column vector such as $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$ in the form $(3, -1)$.

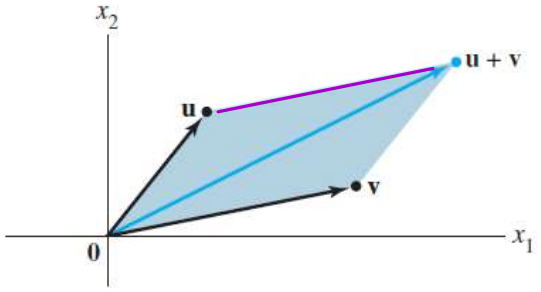
– A column vector $\begin{bmatrix} a \\ b \end{bmatrix}$ in $\mathbb{R}^2$ can be visualized as an arrow from the origin to the (a, b) in $\mathbb{R}^2$.

Ex:

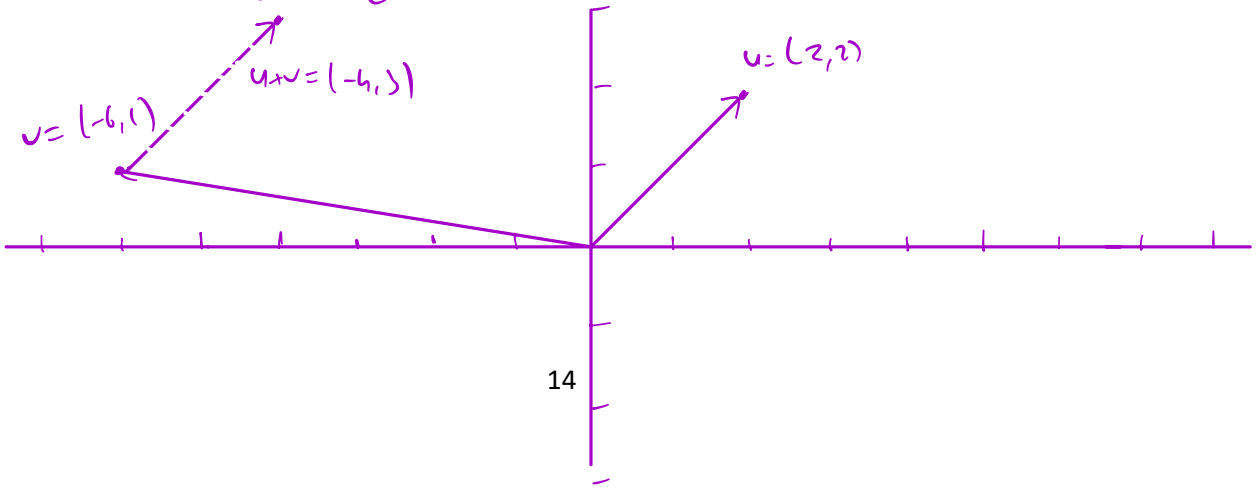


– Parallelogram rule for addition:

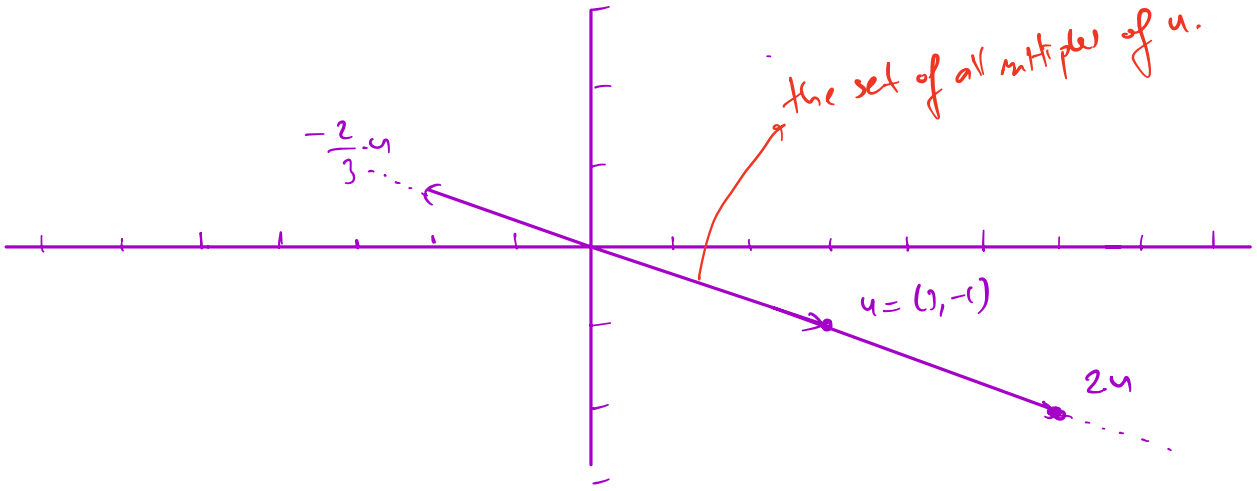
If u and v in $\mathbb{R}^2$ are represented as points in the plane, then $u + v$ corresponds to the fourth vertex of the parallelogram whose other vertices are u, $\mathbf{0}$, and v. ($\mathbf{0}$ corresponds to the zero vector.)



Ex: let $u = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ $v = \begin{bmatrix} -6 \\ 1 \end{bmatrix}$, $u+v = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$



Ex: For $u = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ display $u, 2u, \frac{2}{3}u$.



- Vectors in $\mathbb{R}^3$ are 3x1 column matrices with three entries. For example $a = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$.
- They are represented by points in 3-dimensional coordinate space.
- Vectors in $\mathbb{R}^n$ are written as nx1 column matrices such as $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$.
- The vector whose entries are all zero is called the zero vector.
- Algebraic properties of $\mathbb{R}^n$:

For all u, v, w in $\mathbb{R}^n$ and scalars c and d.

1) $u + v = v + u$
2) $(u+v)+w = u + (v+w)$
3) $u + 0 = 0 + u = u$
4) $u + (-u) = -u + u = 0$
where $-u$ denotes $(-1)u$.

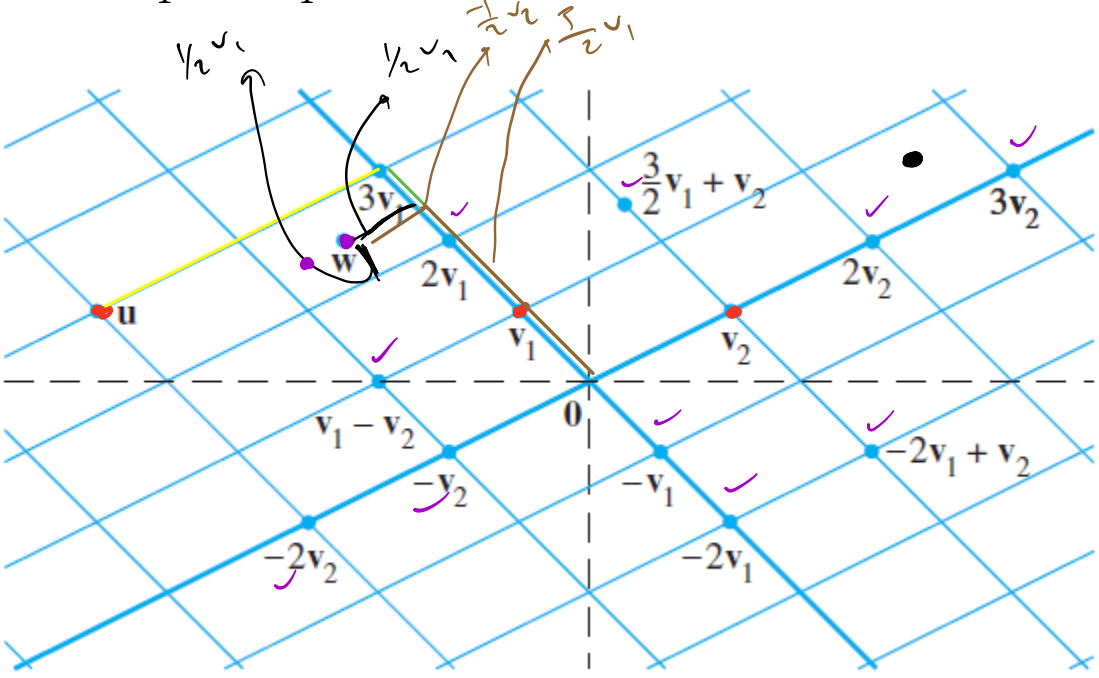
5) $c.(u+v) = cu + cv$
6) $(c+d).u = cu + du$
7) $c.(du) = (cd).u$
8) $1.u = u$

For simplicity of notation, a vector such as $u + (-1)v$ is often written as $u - v$.

Linear combinations:

Given vectors $v_1, v_2, \ldots, v_p$ in $\mathbb{R}^n$ and given scalars $c_1, c_2, \ldots, c_p$ the vector y defined by $y = c_1v_1 + c_2v_2 + \cdots + c_pv_p$ is called a linear combination of $v_1, v_2, \ldots, v_p$ with weights $c_1, c_2, \ldots, c_p$.

Ex: Let $v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Let's identify the linear combinations of v_1 and v_2.



$y = 3v_1 - 2v_2$ y is a linear combination of v_1 and v_2.

$w = \frac{5}{2}v_1 - \frac{1}{2}v_2$ w is a linear combination of v_1 and v_2.

Q: Can I reach any point in $\mathbb{R}^2$ by a linear combination of v_1 and v_2? → YES!

Ex: Let $a_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$, $a_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$, $b = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$

Determine whether b can be generated as a linear combination of a_1 and a_2. [That is, determine whether weights w_1 and w_2 exist such that $x_1 a_1 + x_2 a_2 = b$. If this vector equation has a solution, find it.]

$\text{Span}\{a_1, a_2\}$

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ -2x_1 \\ -5x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 5x_2 \\ 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

vector equation

$$\Rightarrow \begin{bmatrix} x_1 + 2x_2 \\ -2x_1 + 5x_2 \\ -5x_1 + 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix} \Rightarrow \begin{aligned} x_1 + 2x_2 &= 7 \\ -2x_1 + 5x_2 &= 4 \\ -5x_1 + 6x_2 &= -3 \end{aligned}$$

x_1 and x_2 make the vector equation true if and only if x_1 and x_2 satisfy the system.

To solve this system, now reduce the augmented matrix of the system.

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix} \Rightarrow \begin{matrix} 2R_1 + R_2 \to \\ 5R_1 + R_3 \to \end{matrix} \begin{bmatrix} 1 & 2 & 7 \\ 0 & 9 & 18 \\ 0 & 16 & 32 \end{bmatrix} \Rightarrow R_2/9 \to \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 16 & 32 \end{bmatrix}$$

$$\Rightarrow R_3 - 16R_2 \to \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow R_1 - 2R_2 \to \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore, $x_1 = 3$ and $x_2 = 2$.

That is, $3 \cdot \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$

– The discussion and example above is easily modified to establish the following fundamental fact:

A vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{bmatrix} \quad (5)$$

In particular, $\mathbf{b}$ can be generated by a linear combination of $\mathbf{a}_1, \ldots, \mathbf{a}_n$ if and only if there exists a solution to the linear system corresponding to the matrix (5).

– One of the key ideas in linear algebra is to study the set of all vectors that can be generated or written as a linear combination of a fixed set $\{v_1, v_2, \ldots, v_p\}$ of vectors.

– Definition:

If $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $\mathbf{v}_1, \ldots, \mathbf{v}_p$ is denoted by $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned (or generated) by $\mathbf{v}_1, \ldots, \mathbf{v}_p$**. That is, $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is the collection of all vectors that can be written in the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p$$

with $c_1, \ldots, c_p$ scalars.

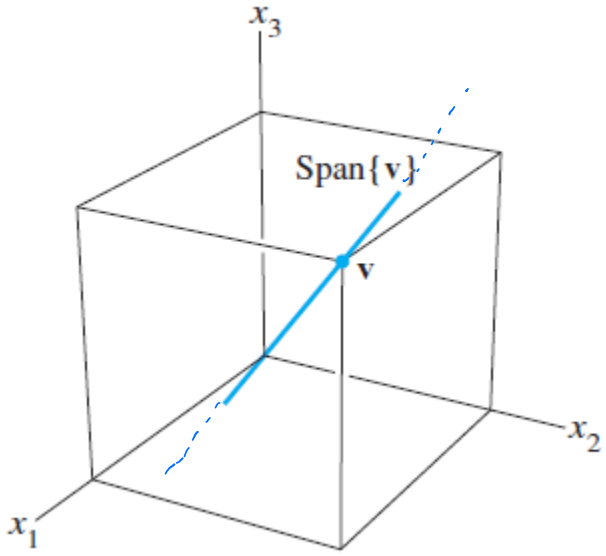
Asking whether a vector b is in $\text{Span}\{v_1, \ldots, v_p\}$ amounts to asking whether the vector equation

$$x_1 v_1 + x_2 v_2 + \cdots + x_p v_p = b$$

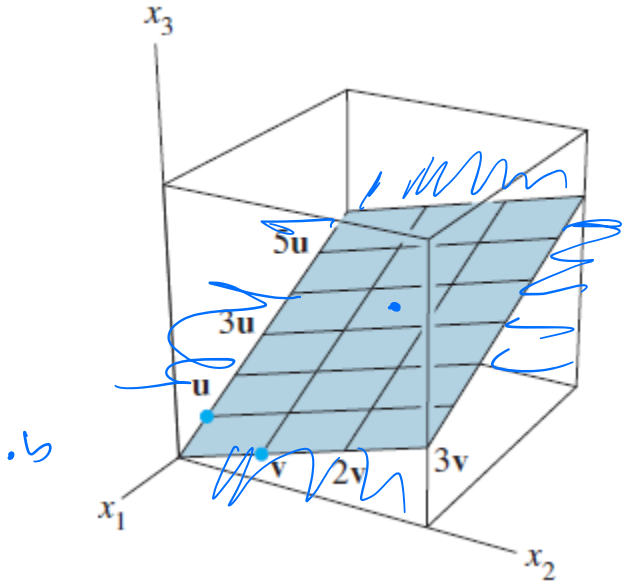
has a solution, or equivalently asking whether the linear system with augmented matrix $[v_1 \ \cdots \ v_p \ \ b]$ has a solution.

$v = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$ Span{v}?

- Geometric Description of Span{v} and Span{u,v}:
- Let v be a nonzero vector in $\mathbb{R}^3$. Then Span{ v } is the set of all scalar multiples of v, which is the set of points on the line in $\mathbb{R}^3$ through v and $\mathbf{0}$.



- If u and v are nonzero vectors in $\mathbb{R}^3$, with v not a multiple of u, then Span{ u, v } is the plane in $\mathbb{R}^3$ that contains u, v and $\mathbf{0}$.



Ex: let $a_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$, $a_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}$ $b = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$.

Then, $\text{Span}\{a_1, a_2\}$ is a plane in $\mathbb{R}^3$.

Is b in that plane?

Does the equation $x_1 a_1 + x_2 a_2 = b$ have a solution?

To answer this, now reduce the augmented matrix $[a_1\ a_2\ b]$:

$$\begin{bmatrix} 1 & 5 & -3 \\ -2 & -13 & 8 \\ 3 & -3 & 1 \end{bmatrix} \to \begin{matrix} R_2 + 2R_1 \to \\ R_3 - 3R_1 \to \end{matrix} \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & -18 & 10 \end{bmatrix}$$

$$\Rightarrow \begin{matrix} \\ \\ R_3 - 6R_2 \end{matrix} \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & 0 & -2 \end{bmatrix} \to \text{no solution} \quad \begin{matrix} 0x_2 = -2 \\ 0 = -2 \ ? \end{matrix}$$

So, b is not in $\text{Span}\{a_1, a_2\}$.